

Thibault de Surrel

Paris, France | thibault.de-surrel@lamsade.dauphine.fr | +33 7 81 69 17 81 | thibaultdesurrel.github.io
linkedin.com/in/thibault-de-surrel/ | github.com/thibaultdesurrel

Education

LAMSADE, Univ. Paris Dauphine - PSL (Paris, France), PhD in Computer Science	2023 – Present
• Title of the PhD: <i>Learning context invariant representations for EEG data.</i> • Supervisors: Florian Yger, Sylvain Chevallier and Fabien Lotte.	
ENS Paris-Saclay (Gif-sur-Yvette, France), Master 2 MVA: Mathematics, Vision, Learning	2022 – 2023
• Master focused on the mathematical aspects of data science, numerical data acquisition, processing and automatic interpretation.	
ENS Paris-Saclay (Gif-sur-Yvette, France), <i>Master 2 Formation pour l'enseignement supérieur</i>	2021 – 2022
• Master of Science in Mathematics teaching. • Agrégation externe de Mathématiques.	
ENSTA Paris (Palaiseau, France), Engineer's Degree - <i>Diplôme d'Ingenieur</i>	2019 – 2023
• ENSTA Paris is amongst the top 5 engineering schools in France. Major in applied mathematics.	
Lycée Claude Fauriel (St-Etienne, France), Preparatory class - <i>Classes préparatoires</i>	2017 – 2019
• Intensive two-year course in mathematics and physics to prepare for National exams to enter top French engineering schools.	

Internships

Research internship , LAMSADE, Univ. Paris Dauphine - PSL (Paris, France)	Apr. 2023 – Sep. 2023
Supervised by: Florian Yger, Sylvain Chevallier and Fabien Lotte	
• Matching trajectories of covariance matrices on the Riemannian manifold of symmetric definite positive (SPD) matrices with application on EEG data. • Building Riemannian versions of the t-SNE and MDS algorithms to visualize high-dimensional SPD matrices.	
Research internship , DataShape team, Inria (Sophia-Antipolis, France)	May 2021 – Aug. 2021
Supervised by: Mathieu Carrière	
• Build a neural network to predict persistence images given a point cloud. • Build a VAE to learn the joint distribution of persistence images and point clouds.	

Publications

A probabilistic view on Riemannian machine learning models for SPD matrices	Oct. 2025
<i>Thibault de Surrel</i> , Florian Yger, Fabien Lotte, and Sylvain Chevallier <i>Geometric Science of Information (GSI 2025)</i>	
Interpretability of Riemannian tools used in brain computer interfaces	Aug. 2025
<i>Thibault de Surrel</i> , Tristan Venot, Marie-Constance Corsi, and Florian Yger <i>IEEE 35th International Workshop on Machine Learning for Signal Processing (MLSP 2025)</i>	
Wrapped Gaussian on the manifold of Symmetric Positive Definite Matrices	Jul. 2025
<i>Thibault de Surrel</i> , Fabien Lotte, Sylvain Chevallier, and Florian Yger <i>ICML 2025</i>	
Geometry-Aware visualization of high dimensional SPD matrices	Feb. 2025
<i>Thibault de Surrel</i> , Sylvain Chevallier, Fabien Lotte and Florian Yger <i>Transactions on Machine Learning Research (TMLR)</i>	

[Averaging trajectories on the manifold of symmetric positive definite matrices](#)

Aug. 2024

Thibault de Surrél, Florian Yger, Sylvain Chevallier and Fabien Lotte

EUSIPCO 2024

[RipsNet: a general architecture for fast and robust estimation of the persistent homology of point clouds](#)

Apr. 2022

Thibault de Surrél, Felix Hensel, Mathieu Carrière, Théo Lacombe, Yuichi Ike, Hiroaki Kurihara, Marc Glisse and Frédéric Chazal

ICLR 2022 Workshop on Geometrical and Topological Representation Learning

Teaching

Linear Model and its Generalization, First year Master, Univ. Paris Dauphine - PSL

2025

Fundamental of ML, Third year Bachelor, Univ. Paris Dauphine - PSL

2024

Mathematical statistics, Third year Bachelor, Univ. Paris Dauphine - PSL

2024-2025

Algebra and numerical methods, Second year Bachelor, Univ. Paris Dauphine - PSL

2024-2025